

Exp No: 12 EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

Github : <https://github.com/Pardhav2006/embedded-system>

Program :

```
#include <REG51.H>

void main(void)
{
    unsigned int num1, num2;
    unsigned long product;
    while(1)
    {
        num1 = 0x2222;
        num2 = 0xB BBB;
        product = (unsigned long)num1 * num2;
        P0 = product & 0xFF;
        P1 = (product >> 8) & 0xFF;
        P2 = (product >> 16) & 0xFF;
        P3 = (product >> 24) & 0xFF;
    }
}
```

C:\Users\yadal\OneDrive\Documents\embedded\EXP 12.uvproj - uVision
 File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers
 Register Value
 Regs
 r0 0x00
 r1 0x00
 r2 0x00
 r3 0x00
 r4 0x00
 r5 0x00
 r6 0x00
 r7 0x00
 Sys
 a 0x00
 b 0x00
 sp_max 0x00
 dptr 0x0000
 PC \$ C:0x0800
 states 389
 sec 0.000422K
 psw 0x00

Disassembly
 3: void main(void)
 4: {
 5: unsigned int num1, num2;
 6: unsigned long product;
 7:
 8: while(1)
 9: {
 10: num1 = 0x2222;
 11: num2 = 0x8888;
 12: product = (unsigned long)num1 * num2;
 13:
 14: P0 = product & 0xFF;
 15: P1 = (product >> 8) & 0xFF;
 16: P2 = (product >> 16) & 0xFF;
 17: P3 = (product >> 24) & 0xFF;
 18: }
 19:
 20: }

EXP 1222 **EXP 12.C**
 1: #include <REG51.H>
 2:
 3: void main(void)
 4: {
 5: unsigned int num1, num2;
 6: unsigned long product;
 7:
 8: while(1)
 9: {
 10: num1 = 0x2222;
 11: num2 = 0x8888;
 12: product = (unsigned long)num1 * num2;
 13:
 14: P0 = product & 0xFF;
 15: P1 = (product >> 8) & 0xFF;
 16: P2 = (product >> 16) & 0xFF;
 17: P3 = (product >> 24) & 0xFF;
 18: }
 19:
 20: }

Parallel Port 0
 Port 0 7 Bits 0
 P0: 0x06 0x06 0x06 0x06
 Pins: 0x06 0x06 0x06 0x06

Parallel Port 1
 Port 1 7 Bits 0
 P1: 0x04 0x04 0x04 0x04
 Pins: 0x04 0x04 0x04 0x04

Parallel Port 2
 Port 2 7 Bits 0
 P2: 0x07 0x07 0x07 0x07
 Pins: 0x07 0x07 0x07 0x07

Parallel Port 3
 Port 3 7 Bits 0
 P3: 0x19 0x19 0x19 0x19
 Pins: 0x19 0x19 0x19 0x19

Command
 Load "C:\\Users\\yadal\\OneDrive\\Documents\\embedded\\Objects\\EXP 12"
 WS 1, 'MAIN
 WS 1, 'num1
 WS 1, 'num2
 WS 1, 'product

Watch 1

Name	Value	Type
MAIN	C:0x0800	Function
num1	0x2222	uint
num2	0x8888	uint
product	0x1907C4D6	ulong
<Enter expression>		

ASH ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE
 Call Stack + Locals Watch 1 Memory 1